

Medical Disease Knowledge Base

1. Diabetes Mellitus

Overview

Diabetes mellitus is a chronic metabolic condition in which blood glucose levels remain higher than normal. This happens because the body does not produce enough insulin, does not use insulin effectively, or both.

Types

The major types of diabetes include Type 1 diabetes, Type 2 diabetes, and gestational diabetes.

Type 1 diabetes occurs when the immune system attacks insulin-producing cells in the pancreas. People with Type 1 diabetes generally require insulin therapy.

Type 2 diabetes occurs when the body becomes resistant to insulin or does not produce sufficient insulin. It is the most common form of diabetes.

Gestational diabetes develops during pregnancy and may increase the mother's and child's risk of developing Type 2 diabetes later in life.

Common Symptoms

Common symptoms may include:

- Frequent urination
- Excessive thirst
- Increased hunger
- Unexplained weight loss
- Fatigue
- Blurred vision
- Slow-healing wounds
- Frequent infections

Some people with Type 2 diabetes may have few or no noticeable symptoms for a long time.

Risk Factors

Risk factors for Type 2 diabetes include obesity, physical inactivity, family history, increasing age, high blood pressure, and certain dietary and lifestyle factors.

Diagnosis

Diabetes can be diagnosed using blood glucose tests such as fasting plasma glucose, oral glucose tolerance testing, or HbA1c testing. A healthcare professional should interpret test results in the appropriate clinical context.

Treatment

Treatment depends on the type and severity of diabetes. Management may include dietary changes, regular physical activity, blood glucose monitoring, oral medications, injectable medications, or insulin.

Prevention

Type 2 diabetes risk can often be reduced through maintaining a healthy weight, regular physical activity, a balanced diet, and regular medical checkups.

2. Hypertension

Overview

Hypertension, commonly called high blood pressure, is a condition in which blood pressure remains consistently elevated. Long-term uncontrolled hypertension can damage the heart, brain, kidneys, eyes, and blood vessels.

Symptoms

Hypertension often does not cause noticeable symptoms. This is why it is sometimes called a "silent" condition.

Very high blood pressure may sometimes be associated with symptoms such as severe headache, chest pain, shortness of breath, visual changes, weakness, or confusion and requires prompt medical assessment.

Risk Factors

Risk factors include increasing age, family history, obesity, high dietary sodium intake, physical inactivity, excessive alcohol consumption, smoking, stress, and certain medical conditions.

Diagnosis

Blood pressure is measured using a blood pressure monitor. A diagnosis generally requires appropriate measurements taken on more than one occasion or through validated monitoring methods.

Complications

Uncontrolled hypertension can increase the risk of:

- Heart attack
- Stroke
- Heart failure
- Kidney disease
- Vision problems
- Blood vessel disease

Treatment

Treatment may include lifestyle modifications and antihypertensive medications. Common medication classes include ACE inhibitors, angiotensin receptor blockers, calcium channel blockers, and diuretics.

Treatment should be individualized by a healthcare professional.

Prevention

Maintaining a healthy weight, exercising regularly, limiting sodium, eating a balanced diet, avoiding tobacco, and monitoring blood pressure can help reduce cardiovascular risk.

3. Asthma

Overview

Asthma is a chronic respiratory condition involving inflammation and narrowing of the airways. Symptoms can vary over time and may become worse when exposed to specific triggers.

Common Symptoms

- Wheezing
- Coughing
- Shortness of breath
- Chest tightness

Symptoms may be worse at night, early in the morning, during exercise, or after exposure to certain triggers.

Common Triggers

Asthma triggers can include allergens, smoke, air pollution, respiratory infections, exercise, cold air, and certain occupational exposures.

Diagnosis

Diagnosis generally involves medical history, physical examination, and lung function testing such as spirometry. Additional testing may sometimes be necessary.

Treatment

Asthma treatment commonly includes inhaled medications. Quick-relief medications may be used for acute symptoms, while controller medications are used regularly to reduce airway inflammation and prevent symptoms.

Prevention

Identifying and avoiding personal triggers, following an asthma action plan, taking prescribed controller medications, and attending regular medical appointments can help control asthma.

4. Pneumonia

Overview

Pneumonia is an infection or inflammation of the lungs that can be caused by bacteria, viruses, fungi, or other organisms.

Symptoms

Common symptoms include:

- Cough
- Fever
- Chills
- Difficulty breathing
- Chest discomfort
- Fatigue
- Production of mucus

Older adults and people with weakened immune systems may have less typical symptoms.

Risk Factors

Risk is higher among older adults, young children, people with chronic diseases, smokers, and individuals with weakened immune systems.

Diagnosis

Diagnosis may involve a physical examination, measurement of oxygen levels, chest imaging, and laboratory tests depending on the suspected cause.

Treatment

Treatment depends on the cause and severity. Bacterial pneumonia may require antibiotics, while viral pneumonia generally does not benefit from antibiotics unless there is a bacterial infection as well.

Severe pneumonia may require hospitalization and oxygen or other supportive treatment.

Prevention

Vaccination, hand hygiene, avoiding smoking, and managing underlying medical conditions can reduce the risk of pneumonia.

5. Migraine

Overview

Migraine is a neurological disorder characterized by recurring attacks of headache and other symptoms. A migraine attack may last several hours to several days.

Symptoms

Symptoms can include:

- Moderate to severe headache
- Throbbing or pulsating pain
- Nausea
- Vomiting
- Sensitivity to light
- Sensitivity to sound

Some people experience an aura before or during a migraine attack. Aura may involve visual disturbances, sensory changes, or speech difficulties.

Triggers

Possible triggers include lack of sleep, stress, hormonal changes, certain foods, dehydration, bright lights, and changes in routine. Triggers vary between individuals.

Diagnosis

Migraine is primarily diagnosed using medical history and clinical evaluation. There is no single blood test that confirms migraine.

Treatment

Treatment may include medications taken during an attack and preventive medications for people with frequent or disabling migraines.

Lifestyle measures such as regular sleep, adequate hydration, regular meals, exercise, and identifying individual triggers may also help.

6. Influenza

Overview

Influenza, commonly known as flu, is a contagious respiratory illness caused by influenza viruses.

Symptoms

Typical symptoms include:

- Fever
- Cough
- Sore throat
- Muscle aches
- Headache
- Fatigue
- Chills

Symptoms often begin suddenly.

Transmission

Influenza can spread when an infected person coughs, sneezes, talks, or through contaminated hands and surfaces.

Treatment

Most people recover with rest, fluids, and supportive care. Antiviral medications may be recommended for certain people, particularly those at increased risk of complications or those with severe illness.

Prevention

Annual influenza vaccination is an important preventive measure. Hand hygiene, covering coughs and sneezes, and staying home when sick can also reduce transmission.

7. Anemia

Overview

Anemia is a condition in which the blood does not have enough healthy red blood cells or sufficient hemoglobin to adequately carry oxygen to tissues.

Symptoms

Common symptoms include:

- Fatigue
- Weakness
- Dizziness
- Shortness of breath
- Pale skin
- Headache
- Difficulty concentrating

Causes

Anemia can result from iron deficiency, vitamin deficiencies, blood loss, chronic diseases, inherited conditions, or problems with red blood cell production.

Diagnosis

Diagnosis generally involves a complete blood count and may include additional tests such as iron studies, vitamin levels, or tests for blood loss depending on the suspected cause.

Treatment

Treatment depends on the underlying cause. Iron-deficiency anemia may require dietary changes and iron supplementation, while other types require different treatments.

It is important to identify the cause rather than treating anemia solely based on the hemoglobin level.

8. Osteoarthritis

Overview

Osteoarthritis is a common joint disorder involving changes to cartilage and other structures within a joint. It commonly affects the knees, hips, hands, and spine.

Symptoms

Symptoms may include:

- Joint pain
- Stiffness
- Reduced range of motion
- Joint swelling
- Tenderness
- Pain that worsens with activity

Morning stiffness is usually relatively brief compared with inflammatory forms of arthritis.

Risk Factors

Risk factors include increasing age, previous joint injury, obesity, repetitive joint stress, and genetic factors.

Diagnosis

Diagnosis is generally based on symptoms, physical examination, and sometimes imaging.

Treatment

Treatment may include exercise, physical therapy, weight management when appropriate, pain-management strategies, assistive devices, and medications. In severe cases, joint replacement surgery may be considered.

9. Rheumatoid Arthritis

Overview

Rheumatoid arthritis is a chronic autoimmune inflammatory disease that primarily affects joints. It occurs when the immune system mistakenly attacks tissues in the joints.

Symptoms

Common symptoms include:

- Joint pain

- Joint swelling
- Morning stiffness
- Warmth around affected joints
- Fatigue
- Reduced joint function

Symptoms often affect joints symmetrically, although patterns can vary.

Diagnosis

Diagnosis may involve medical history, physical examination, blood tests such as rheumatoid factor and anti-CCP antibodies, inflammatory markers, and imaging.

Treatment

Treatment aims to reduce inflammation, prevent joint damage, maintain function, and improve quality of life. Disease-modifying antirheumatic drugs, biologic medications, targeted therapies, physical therapy, and other treatments may be used.

Importance of Early Treatment

Early diagnosis and appropriate treatment can help reduce the risk of permanent joint damage and disability.

10. Tuberculosis

Overview

Tuberculosis (TB) is an infectious disease caused by bacteria of the *Mycobacterium tuberculosis* complex. It most commonly affects the lungs but can affect other organs.

Symptoms of Pulmonary TB

Symptoms may include:

- Persistent cough
- Fever
- Night sweats
- Weight loss
- Fatigue
- Chest pain
- Coughing up blood in some cases

Transmission

Pulmonary TB can spread through the air when a person with infectious TB coughs, speaks, or sings.

Diagnosis

Diagnosis may involve chest imaging, sputum testing, molecular tests, and other laboratory investigations.

Treatment

Active TB is treated with a combination of antibiotics taken for several months. The exact regimen depends on factors such as drug susceptibility and the location of infection.

Completing the prescribed treatment is important because incomplete treatment can contribute to drug-resistant TB.

Prevention

Early diagnosis and appropriate treatment of infectious TB help reduce transmission. Vaccination with BCG is used in many countries, particularly to protect children against severe forms of TB.

Medical Safety Notice

This information is intended for educational and information-retrieval purposes only. It is not a substitute for diagnosis, medical advice, or treatment from a qualified healthcare professional.

Medical symptoms can have many different causes. Users should not make treatment decisions solely from information retrieved from this knowledge base.

For severe or rapidly worsening symptoms, users should seek appropriate medical care promptly.